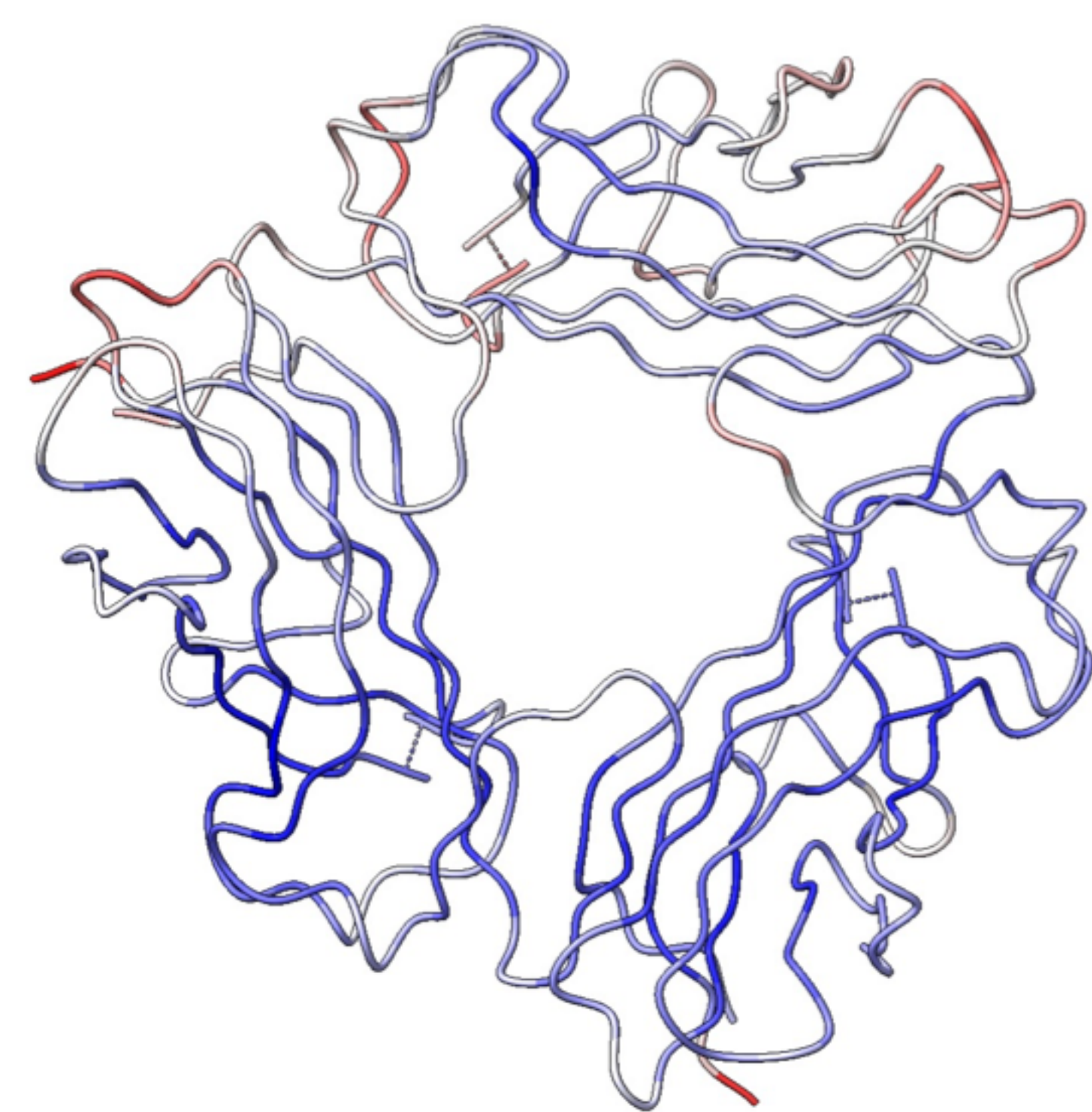


## Extract Data from Proteins

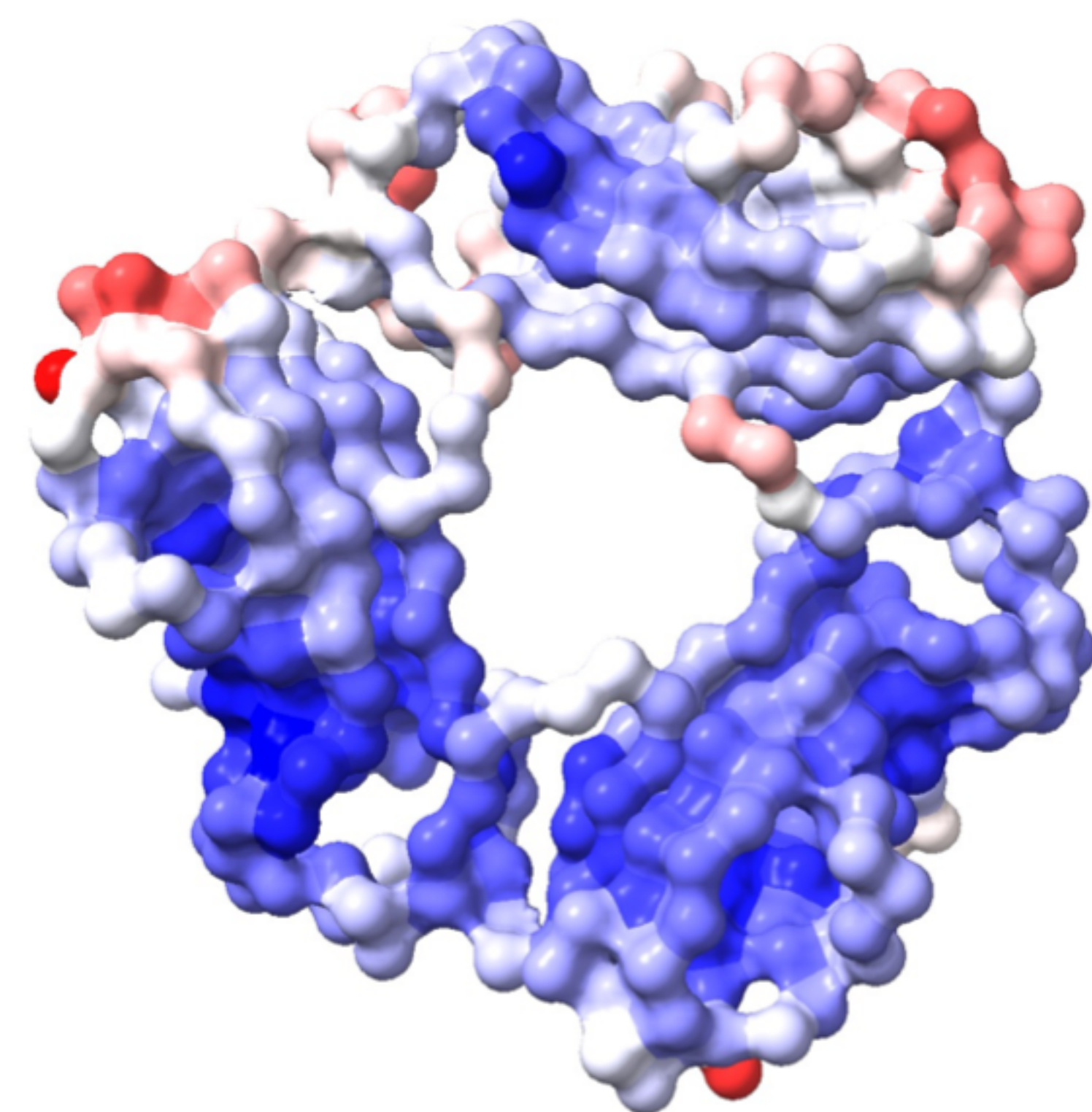
- Protein structures
- C $\alpha$  atoms coordinates
- B-factor values



## Local Atom Group Construction

Atom-specific local neighborhoods

Nearby atoms are selected within multiple spatial cutoff radius



## Commutative Algebra Featurization

Construct a filtration of simplicial complexes with atomic local neighborhoods

F-vector descriptors extracted from simplicial complexes

## Construct Atomic Features

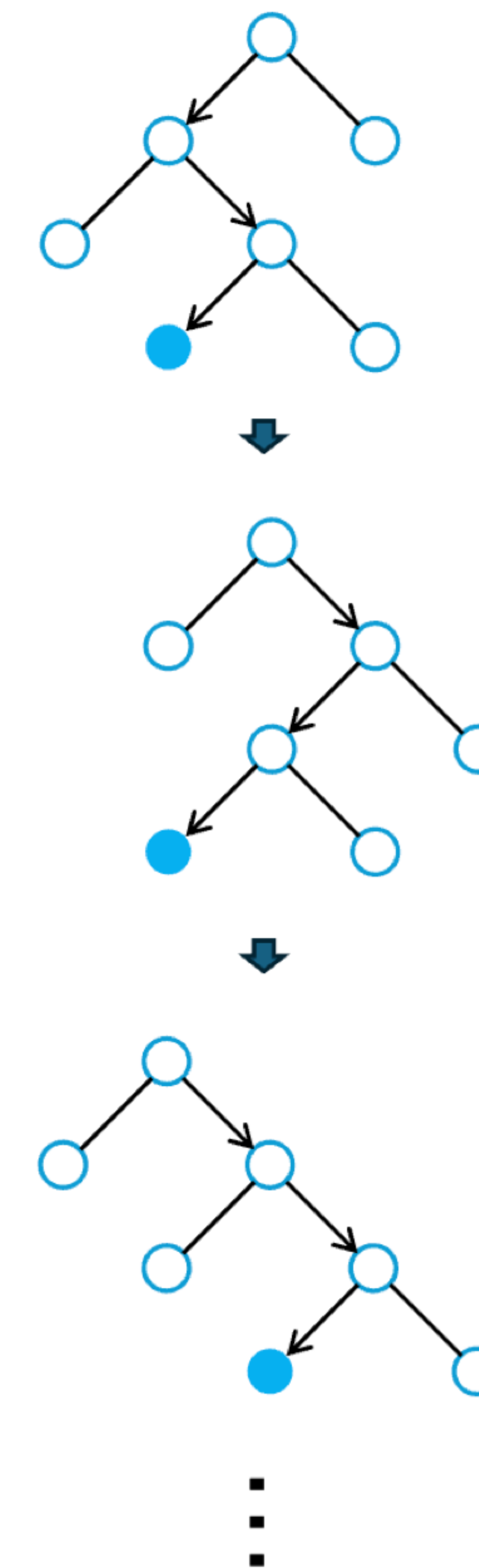
Commutative algebra structural descriptors



Physicochemical descriptors (global + local residue features)

Linear-model predictions

## Machine Learning Models



Blind Predictions

Atom-level cross-validation

Protein-level cross-validation

Leave-one-protein-out validation